Arista Access Node WFD-ACC-31 QoS Congestion Event Resolution SOP

Node: WFD-ACC-31 | **Vendor:** Arista | **Severity:** Warning | **Alarm Code:** ALARM_3357

Description

The QoS congestion event failure domain on Arista access nodes is primarily caused by the combination of microbursts, interface tail drops, and QoS queue starvation. This phenomenon is exacerbated by the high-speed nature of modern networking environments, where bursty traffic patterns can overwhelm the node's forwarding capacity. In the context of Arista access nodes, this is often due to the node's role as a key aggregation point for multiple subscriber connections, where each subscriber's traffic can contribute to the overall congestion burden. Furthermore, the node's QoS configuration, including the scheduling and queuing policies, can influence the likelihood and severity of congestion events. In this SOP, we will guide the resolution of QoS congestion events on the Arista access node WFD-ACC-31, focusing on the identification of root causes, execution of diagnostic commands, implementation of resolution steps, verification of fixes, and rollback procedures as necessary.

Pre-requisite Checks

- item: Verify redundant link status on WFD-ACC-31, description: Ensure that all redundant links are operational and not causing any issues

- item: Check Whitefield POP environmental conditions, description: Verify that the POP environment is within normal operating temperature and humidity ranges

- item: Confirm no ongoing maintenance or upgrades on WFD-ACC-31, description: Ensure that no maintenance or upgrades are scheduled or in progress on the node

Potential Root Causes

Cause	Probability
Microbursts and interface tail drops	High
QoS queue starvation	Medium
Misconfigured QoS policies	Low

Diagnostic Commands & Expected Outputs

```
Command: show interface Ethernet1/1, Expected Output: Ethernet1/1 is up, line protocol is up, Description: Verify the status of interface Ethernet1/1 to determine if it is up and operational

Command: show queue statistics, Expected Output: Queue statistics: Drops: 0, Overflows: 0, Description: Check the queue statistics to identify any drops or overflows, which may indicate congestion
```

```
Command: show qos policy, Expected Output: Policy name:
WFD-ACC-31-qos, Description: Verify the QoS policy configuration to
ensure it is properly set up and not contributing to congestion
```

Resolution Steps

Step 1: Step: Check and adjust QoS policy, Command: configure terminal policy-map WFD-ACC-31-qos class class-default bandwidth 100 exit, Description: Adjust the QoS policy to increase the bandwidth allocation to the class-default class to alleviate congestion

Step 2: Step: Verify interface configuration, Command: show interface Ethernet1/1, Description: Verify that the interface configuration is correct and not contributing to congestion

Step 3: Step: Check and adjust queue configuration, Command: configure terminal interface Ethernet1/1 queue-limit 1000 exit, Description: Adjust the queue configuration to increase the queue limit to alleviate congestion

Post-Check Verification

```
Command: show queue statistics, Expected Output: Queue statistics:
Drops: 0, Overflows: 0, Description: Verify that the queue statistics
have returned to normal, indicating that congestion has been
alleviated

Command: show qos policy, Expected Output: Policy name:
WFD-ACC-31-qos, Description: Verify that the QoS policy configuration
has been successfully adjusted
```

Roll-back Procedures

- Step: Restore original QoS policy, Command: configure terminal policy-map WFD-ACC-31-qos class class-default bandwidth 50 exit, Description: Restore the original QoS policy configuration to its previous state

- Step: Restore original interface configuration, Command: configure terminal interface Ethernet1/1 queue-limit 500 exit, Description: Restore the original interface configuration to its previous state